



Nguyen Hoang Viet

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SUMMARY

My research lies at the intersection of statistics and machine learning, where I seek to understand the behavior of machine learning models from a rigorous theoretical perspective. My recent work focuses on parameter convergence rates in Mixture-of-Experts models. I am also interested in quantum computing, especially quantum optimization and quantum machine learning. My long-term goal is to develop statistical methodologies that enhance the reliability and interpretability of machine learning systems across both classical and quantum computing platforms.

EDUCATION

2019 - 2022 **High School for Gifted Students, Hanoi University of Science (HUS)** Vietnam
High School Diploma (Specialized in Mathematics)
2022 - 2026 **Hanoi University of Science and Technology (HUST)** Hanoi, Vietnam
(Expected) BSc in Computer Science (Talented Program) (CPA: 3.90/4.0 - top 1% of the cohort)

RESEARCH EXPERIENCE

Independent Researcher Oct 2025 - present
– Supervisor: [Prof. Nhat Ho](#), The University of Texas at Austin
– Focused on the theoretical analysis of Mixture-of-Experts models, with particular emphasis on parameter convergence rates
– Conducted research on medical image retrieval methods, focusing on colorectal cancer detection
Undergraduate Research Assistant | NetLab - HUST Jul 2024 - Sep 2025
– Supervisor: [Prof. Trinh Chien](#), Hanoi University of Science and Technology
– Formulated the communication channel estimation problem using rigorous mathematical analysis and introduced an evolutionary algorithm to improve channel quality
– Conducted research using the D-Wave quantum computer to solve multi-objective optimization problems via quantum annealing

JOURNAL PUBLICATIONS

[Advanced Quantum Annealing for the Bi-Objective Traveling Thief Problem: An \$\varepsilon\$ -Constraint-Based Approach](#)
Viet Nguyen, Nguyen Xuan Tung, Trinh Van Chien, Won-joo Hwang
IEEE Transactions on Quantum Engineering
[Improved Differential Evolution for Enhancing the Aggregated Channel Estimation of RIS-Aided Cell-Free Massive MIMO](#)
Trinh Van Chien, **Viet Nguyen**, Symeon Chatzinotas, Lajos Hanzo
IEEE Transactions on Vehicular Technology 99, pp. 1-6

CONFERENCE PUBLICATIONS

[A Mixture of Experts Vision Transformer for High-Fidelity Surface Code Decoding](#)

Viet Nguyen, Manh Hung Nguyen, Hoang Ta, Van Khu Vu, Yeow Meng Chee

IEEE International Symposium on Information Theory (ISIT 2026)

PREPRINTS

[Rethinking Multinomial Logistic Mixture of Experts with Sigmoid Gating Function](#)

Tuan Minh Pham*, Thinh Cao*, **Viet Nguyen***, Huy Nguyen, Nhat Ho**, Alessandro Rinaldo**

Under Review

[A Statistical Theory of Gated Attention through the Lens of Hierarchical Mixture of Experts](#)

Viet Nguyen*, Tuan Minh Pham*, Thinh Cao*, Huy Nguyen, Nhat Ho**, Alessandro Rinaldo**

Under Review

ACHIEVEMENT

2020	Vietnam Mathematical Olympiad	(Second Prize – ranked 13th)
2023	Vietnam Math Competition for University student	(First Prize – ranked 4th)
2025	Vietnam Math Competition for University student	(First Prize – ranked 8th)

LANGUAGES

Vietnamese	Native	
English	IELTS Overall 7.0	(L – R – W – S: 6.5 – 8.5 – 7.0 – 6.0)

REFERENCES

Prof. Nhat Ho	The University of Texas at Austin
Prof. Trinh Chien	Hanoi University of Science and Technology